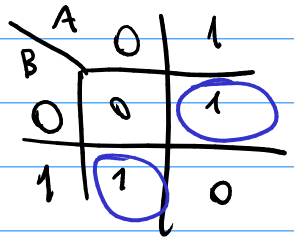
Ejercicio 1: Half-subtractor

a. Tabla de verdad

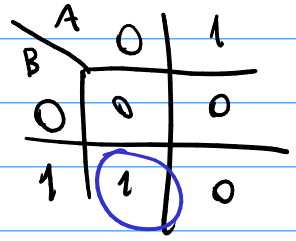
A	B	S	B_{out}	
0	0	0	0	→ 0 − 0 = 0
0	1	1	1	→ 0 − 1 = 1 (con borrow)
1	0	1	0	→ 1 − 0 = 1
1	1	0	0	→ 1 − 1 = 0

b. K-Maps

Para S:



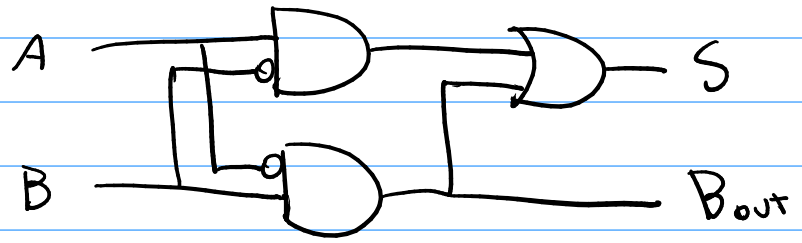
Para B_{out}:



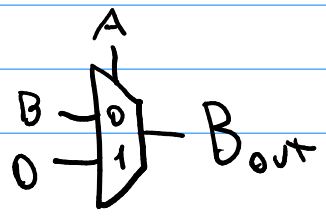
$$\Rightarrow S = A\bar{B} + \bar{A}B \quad , \quad B_{out} = \bar{A}B$$

↖ de hecho, esto es A xor B

c. Circuito



f. B_{out} usando un mux 2:1



g. S usando un mux 4:1

